

SFT V11.6 Addendum to V11.5

Boundary-Time Duality and the History Axis

One tick, two projections: proper-time advance and boundary-capacity growth

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Status. This is a conceptual and mathematical addendum to *SFT V11.5: The One-Speed Universe*. It sharpens the one-speed principle by introducing a history-index formalism. It does not claim a completed theorem of quantum gravity; it states the SFT interpretation in a form that can be used by later derivations.

Abstract. V11.5 interpreted the speed of light as the universal causal processing rate: one Planck length per one Planck time, one spatial operation per one temporal operation. This addendum adds the missing conceptual bridge. A tick does not separately advance time and write boundary area. A tick is an irreversible history increment. The local face of that increment is proper-time or phase advance; the holographic face is boundary-capacity growth. Boundary area and time are not dimensionally identical, but in SFT they are dual measures of the same count variable: accumulated irreversible history.

1. The Question Left Open by V11.5

V11.5 stated the one-speed principle: every physical process has a fixed c-normalized causal budget. A photon expresses that budget as null spatial propagation. A massive particle expresses it as proper-time advance and internal phase cycling. Motion changes the projection of the budget; it does not create motion from stillness.

That note still left one conceptual seam. SFT says a particle tick advances time and also adds one unit of boundary capacity. Why should those two statements be tied together?

The answer proposed here is that they are not two independent consequences. They are two projections of one event:

one irreversible tick = one history increment

The particle experiences that increment as proper-time or phase advance. The universe accounts for it as boundary-capacity growth.

2. The History Index

Introduce a dimensionless history index N . For a particle or field mode i , the increment dN_i counts complete irreversible maintenance updates. For a massive rest-clock this is the Compton phase count:

$$dN_i = \nu_i d\tau_i = \frac{E_i}{h_P} d\tau_i$$

History count from local proper-time advance.

For a rest particle with $E_i = m_i c^2$,

$$\nu_{C,i} = \frac{m_i c^2}{h_P}, \quad T_{C,i} = \frac{h_P}{m_i c^2}$$

Thus one complete Compton cycle gives

$$dN_i = \frac{d\tau_i}{T_{C,i}} = \frac{d\phi_i}{2\pi}$$

Guardrail. Planck time and Compton time are not being identified. The Planck time is the causal resolution scale. A Compton cycle is a particle-specific phase period built from many elementary causal operations. SFT assigns the irreversible history count to complete maintenance cycles, while the boundary capacity unit remains Planck-area scaled.

3. Boundary Capacity as the Holographic Face of History

Let the elementary boundary-capacity increment associated with one history count be

$$dA_i = \xi \ell_P^2 dN_i$$

Here ξ is a convention factor. The simplest SFT convention uses $\xi = 1$; Bekenstein-Hawking entropy conventions may absorb factors such as 4 or $\ln 2$ into the definition of one capacity unit. The addendum keeps ξ explicit so the physical statement is independent of normalization.

Combining local phase count and boundary capacity gives the boundary-time duality identity:

$$\frac{dA_i}{\xi \ell_P^2} = \frac{d\tau_i}{T_{C,i}} = \frac{d\phi_i}{2\pi} = dN_i$$

Boundary area and proper-time phase are dual measures of the same history increment.

This is the mathematical form of the sentence: the tick does not advance time and then separately write to the boundary. The tick is one irreversible history increment. Time is its local face. Boundary area is its holographic face.

Taking a proper-time derivative recovers the area-demand rate already isolated in the Planck-area accounting note:

$$\frac{dA_i}{d\tau_i} = \xi \ell_P^2 \frac{2m_i c^2}{h_P} = \frac{\xi G m_i}{2\pi c}$$

The result is compact: mass fixes how quickly a persistent process demands holographic capacity. The Planck area supplies the unit; the Compton frequency supplies the rate.

4. The Particle Never Returns to the Same Event

The visual origin of this addendum is the helical worldline. A particle may repeat a spatial projection - circling, oscillating, returning near the same x, y, z - but it never returns to the same spacetime event.

$$X_n^\mu = (ct_n, \mathbf{x}_n), \quad X_{n+1}^\mu = (ct_{n+1}, \mathbf{x}_{n+1})$$

Even if the spatial projection repeats,

$$\mathbf{x}_{n+1} = \mathbf{x}_n \text{ / } \Rightarrow \quad X_{n+1}^\mu \neq X_n^\mu$$

because

$$t_{n+1} > t_n$$

The old spatial location can be revisited. The old four-dimensional event cannot. The previous event is history. In SFT language, it has already been written to the boundary-accounting ledger.

Physics-safe wording. This note does not deny 3D space as a useful slice. It states the relativistic fact that a 3D position is not a full event. The full event includes time, and the same spatial coordinates at a later time correspond to a new history entry.

5. The One-Speed Principle in History Coordinates

The fixed c-budget can now be written in a safer form than the phrase "everything moves at c." Define

$$\beta_{\text{space}} = \frac{v}{c}, \quad \beta_{\text{history}} = \frac{d\tau}{dt}$$

For inertial special-relativistic motion,

$$\beta_{\text{space}}^2 + \beta_{\text{history}}^2 = 1$$

or equivalently

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}$$

The coordinate-observed rest-clock rate is then

$$\frac{dN_i}{dt} = \nu_{C,i} \frac{d\tau_i}{dt} = \frac{m_i c^2}{h_p} \sqrt{1 - \frac{v^2}{c^2}}$$

This equation does not say the particle loses rest mass when it moves. It says the internal phase count registered per unit external coordinate time is time-dilated. The rest-clock still advances at its invariant rate per unit proper time.

In a static gravitational field, the same history-rate idea applies with the metric redshift factor. For a static clock outside a Schwarzschild mass,

$$\frac{dN_i}{dt} = \nu_{C,i} \sqrt{1 - \frac{2GM}{c^2 r}}$$

This is a coordinate statement for a static clock. It does not imply that local physics stops for an infalling observer at a horizon. It states how the history-count rate projects into the chosen external time coordinate.

6. Boundary Growth as Global History Accumulation

Let I denote boundary capacity measured in capacity units:

$$I = \frac{A}{\xi \ell_p^2}$$

The boundary-time duality says that global capacity growth is the sum of local history increments:

$$dI = \sum_i dN_i$$

and therefore

$$\frac{dl}{dt} = \sum_i \frac{E_i}{h_P} \frac{d\tau_i}{dt}$$

In the homogeneous cosmological limit this becomes the SFT master identity:

$$H = \dot{\frac{S}{I}}$$

The addendum clarifies what this identity means. Expansion and information accumulation are not two unrelated rates set equal by a postulate. They are two macroscopic measurements of accumulated irreversible history. Locally the history index is time/phase advance. Holographically it is boundary capacity. Cosmologically it is expansion.

7. One Tick, Two Faces

Face of the update	Mathematical measure	SFT meaning
Local phase	$d\phi / 2\pi$	The particle completes a maintenance cycle.
Proper time	$d\tau / T_C$	The local clock advances by one count.
Boundary capacity	$dA / (\xi l_P^2)$	The holographic ledger gains one capacity unit.
History index	dN	The invariant count behind the three descriptions.

The most compact statement is therefore:

$$\text{time advance} \equiv \text{history count} \equiv \text{boundary accounting}$$

The equivalence is not dimensional equality. It is equality after projection onto the common count variable N .

8. What This Addendum Establishes

1. The boundary write and local time advance are dual projections of one irreversible history increment.
2. The correct invariant variable is the history index N , not area alone and not time alone.
3. For a massive rest-clock, $dA/(\xi l_P^2) = d\tau/T_C = d\phi/(2\pi) = dN$.
4. A particle may revisit the same spatial coordinates, but it cannot revisit the same spacetime event; the old event is already history.
5. The one-speed principle becomes a history-budget principle: spatial motion and proper-time/history advance are projections of one fixed causal budget.
6. Global expansion is the boundary-scale accounting of the sum of all local history increments.

Open status. This addendum is a structural clarification, not a completed derivation of microscopic spacetime. The remaining gates are: normalization of ξ , the full covariant definition of the history current, integration with the repaired action, and the relationship between Compton-cycle history counts and Planck-scale causal operations.

Closing statement

A tick is not something that happens in time and then writes to a boundary. A tick is the act by which time and boundary are jointly updated. Locally, the update is a clock tick. Holographically, it is a capacity increment. Cosmologically, the accumulation of such increments is expansion. One operation, one history increment, two projections.

$$\frac{dA}{\xi \ell_P^2} = dN = \frac{d\tau}{T_C}$$

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